Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

PROJECT PROPOSAL

RUET Chottor

A Unified Web Platform for Student Communication
and Campus Services

Submitted by	**Submitted to**
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1. Introduction

RUET Chottor is a proposed web platform that brings the everyday communication and campus needs of RUET students into one place. The word *chottor* means a campus square: the open space where students meet, share news, and find one another. The platform aims to be the digital version of that space.

Today, much of student life at RUET is coordinated through a scattered mix of Facebook groups, WhatsApp groups, notice boards, and word of mouth. RUET Chottor will offer a single, dedicated platform for finding people, talking to them in real time, and using common campus services such as event listings, a lost-and-found board, a student marketplace, and bus schedules.

2. Background and Problem Statement

Through my own experience as a student, and from talking with classmates, I have noticed several recurring difficulties:

- **Finding contacts is hard.** There is no reliable place to look up the contact details of alumni, teachers, or even fellow students from another section or series.
- **Communication is fragmented.** Students create a new Facebook or WhatsApp group for almost every purpose, including small groups such as short-semester classmates. Information gets lost across dozens of groups, and students who are not added simply miss it.
- **Campus services have no common home.** Club events, campus news, lost items, second-hand books and equipment, and bus timings are each shared through different channels, if at all.
- **Misinformation spreads easily.** Open social media groups have little moderation, so rumours and spam can spread quickly.

Google Classroom already serves course management well, and RUET Chottor does not try to replace it. The gap is everything outside the classroom: people, communication, and campus life.

3. Objectives

The main objectives of this project are:

- To build a searchable directory of RUET students, alumni, and teachers.
- To provide real-time one-to-one and group messaging for the RUET community.
- To give clubs a simple way to announce and manage events, and students a simple way to find them.
- To provide a moderated news section, a lost-and-found board, and a campus marketplace.
- To publish RUET bus schedules and, where feasible, live bus locations.
- To keep the platform safe through verified sign-up and an admin moderation panel.

- To release the project as open-source software that other students can maintain and extend.

4. Scope of the Project

The platform is divided into independent modules. Each module can be developed, tested, and released on its own. The modules planned for the platform are listed below.

Module	Description
Contact Directory	Search students, alumni, and teachers by name, department, series, or roll. Users control which contact details are visible.
Real-time Messaging	One-to-one chat and group chat, including groups for a series, section, or short semester.
Club & Event Management	Clubs maintain a page, post events, and manage registrations. Students browse upcoming events in one calendar.
News	Campus news and announcements, posted by verified accounts and reviewed by moderators.
Lost & Found	Report a lost or found item with a description, photo, and location on campus.
Campus Marketplace	Buy and sell used books, lab equipment, and other items within the RUET community.
Bus Schedule & Tracker	Up-to-date bus routes and timings, with live location tracking as a later addition.
Admin Panel	Tools for moderators to review reports, remove harmful or false content, and suspend accounts that spam or abuse the platform.

Course management, grading, and other academic functions are outside the scope of this project.

5. User Accounts and Security

Since the platform is meant only for the RUET community, account creation needs to be simple but verified.

5.1 Registration

A new user signs up with their name, roll number, registration number, email address, phone number, date of birth, and a password.

5.2 Login

Users log in with their roll number and password. Two-factor authentication is then completed with a one-time code sent to the user's email or phone.

5.3 Data Protection

- Passwords will be stored only as salted hashes, never in plain text.
- Personal details such as phone numbers will be visible only to the extent each user allows.

- Role-based access will separate regular users, club managers, and administrators.

6. Usability and User Guidance

The platform will not be promoted through marketing, so it has to explain itself. Every main screen will include short descriptions of what it does, and small information (i) icons will explain individual options when hovered or tapped. First-time users will see a brief introduction the first time they open each section.

The visual design will be clean and minimal. Function and ease of use will take priority over decoration, and the layout will work on both mobile phones and computers, since most students use their phones.

7. Development Methodology

- **Incremental releases.** Features will be introduced one at a time instead of all at once. Each release can be tested by real users before the next one begins.
- **Modular codebase.** Each section will be a separate module with clear boundaries, so that one feature can change without breaking the others.
- **Open source.** The source code will be public, with structured documentation covering setup, architecture, and contribution guidelines. This lets future RUET students take over and continue the work.
- **Initial data.** During development, the platform will use dummy data and publicly available information. No real personal data will be used until the platform is ready.

8. Proposed Technology

The project will use stable, widely used, open-source tools with active communities. The choices below are tentative and may be revised during development.

Layer	Proposed Choice
Frontend	React with Next.js
Backend	Node.js (Express or NestJS)
Database	PostgreSQL
Real-time communication	Socket.IO
Authentication	Session/JWT based login, with one-time codes via email (and SMS where available)
Maps (bus tracker)	Leaflet with OpenStreetMap
Version control	Git and GitHub

9. Development Plan

The work is planned in seven phases over roughly sixteen weeks. The admin panel will be built alongside the other modules, so moderation is in place from the first release.

Phase	Work	Duration
1	Requirement analysis, system design, database design	2 weeks
2	Sign-up, login, two-factor authentication, user profiles, contact directory	3 weeks
3	Real-time messaging and group chat	3 weeks
4	News section, club and event management	2 weeks
5	Lost & found board and campus marketplace	2 weeks
6	Bus schedule, followed by live tracking if feasible	2 weeks
7	Testing, bug fixing, documentation, and final report	2 weeks

10. Challenges and Considerations

- **Live bus tracking** requires a location source on each bus, such as a phone carried by the driver or a volunteer. The first version will therefore provide schedules, and live tracking will be added once a practical location source is arranged.
- **SMS verification** has a running cost. Email will be the default channel for one-time codes, with SMS added if a free or sponsored gateway becomes available.
- **Moderation** depends on people as well as tools. A small team of trusted moderators will be needed to keep the platform clean once it has real users.
- **Privacy** is important because the directory contains personal details. Users will decide what is visible, and the platform will collect only the information it needs.

11. Expected Outcomes

- A working, deployed web platform with the modules described in Section 4.
- Fewer scattered social media groups, with one trusted place for RUET communication.
- Easier contact between students, alumni, and teachers.
- A public, documented codebase that future students can maintain and extend.

12. Conclusion

RUET Chottor addresses a real and everyday problem for RUET students: the people and services of campus life are spread across too many places. By bringing them together in one moderated, easy-to-use platform, and building it in small, well-documented steps, the project aims to become a lasting resource for the RUET community. I respectfully request your approval to begin work on this project, and I would welcome any suggestions you may have.

Supervisor's Remarks	
Signature: _____	Date: _____